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INTRODUCTION

Road safety is a vital public concern. Every day, multiple factors—such as lighting, weather, time of day, and road characteristics—affect accident likelihood.

The purpose of this project is to perform Exploratory Data Analysis (EDA) and visualization on a dataset of synthetic road accidents to identify patterns, correlations, and high-risk factors.

This project applies visualization tools to reveal which conditions are most likely to lead to accidents and how environmental or infrastructural aspects contribute. Instead of predictive modeling, the emphasis is on *visual insights* and *data-driven interpretation*.

Objectives:

- Examine accident distribution across road types (rural, highway, urban).
- Analyze factors like lighting, weather, curvature, and speed limits.
- Investigate the influence of time of day and seasons on accident risk.
- Discover correlation between road conditions and accident severity.

METHODOLOGY

1. Problem Definition

To identify the key road, environmental, and situational factors contributing to higher accident risk through visualization and descriptive analysis.

2. Data Collection

Dataset: [synthetic road accidents 2k.csv](#)

Features include:

- Road Type (rural, urban, highway)
- Number of Lanes
- Curvature
- Speed Limit
- Lighting Conditions
- Weather
- Public Road Status
- Time of Day
- Holiday Indicator
- School Season Indicator
- Number of Reported Accidents
- Accident Risk (0–1 scale)

3. Data Preprocessing

Performed using pandas:

- Loaded dataset from CSV
- Checked data info and null values
- Encoded categorical variables as needed
- Created visual-friendly columns (e.g., simplified labels)
- Verified numeric types for correlation and plots

4. Exploratory Data Analysis

EDA methods applied:

- Univariate: Frequency of accident risk by one feature (e.g., weather)
- Bivariate: Relationships between two variables (e.g., curvature vs risk)
- Multivariate: Visualizing multiple factors together using heatmaps

5. Visualization Tools

Libraries used:

- Matplotlib – plotting
- Seaborn – interactive/statistical visualization
- Squarify – treemap representations
- Used plots:
 - i. Countplots
 - ii. Box plots
 - iii. KDE Plots
 - iv. Heatmaps
 - v. Line Charts

FINDING AND ANALYSIS

1. Dataset Overview

Dataset includes 2000+ rows of road accident records across multiple road types. Variables span structural (road type, curvature) and situational (weather, lighting) factors.

2. Observations

- Highways generally show higher speed limits and risk scores.
- Curvature increases accident risk for all road categories.
- Nighttime and foggy/rainy conditions correspond to higher accident probabilities.
- School season days tend to have slightly more accidents (possible increased traffic).

3. Risk Patterns

Accident risk correlates moderately with:

- Curvature
- Lighting
- Weather conditions

4. Statistical Summary

Correlation coefficients highlight curvature and lighting as the top numerical influencers of accident risk.

CODE AND OUTCOMES

Import Required Libraries:

```
import numpy as np
import pandas as pd
import seaborn as sb
import matplotlib as plt
```

Load dataset:

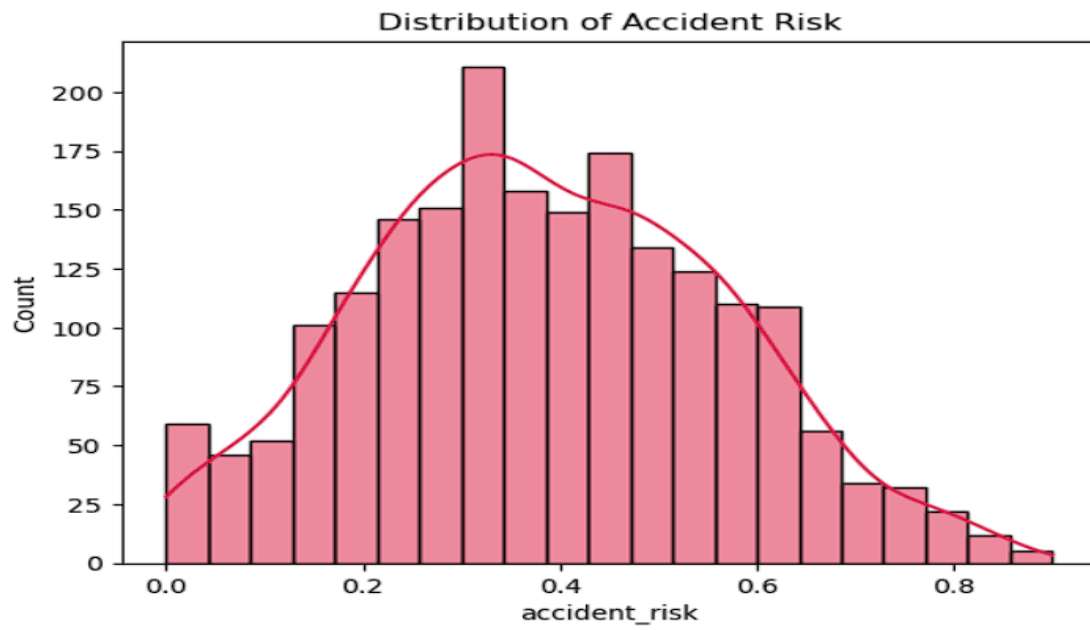
```
df=pd.read_csv(r"C:\Users\Debangi Mondal\Desktop\Python\BCT\Project\synthetic_road_accidents_2k.csv")
print(df)
```

Data Cleaning:

```
df.dropna(inplace=True)
```

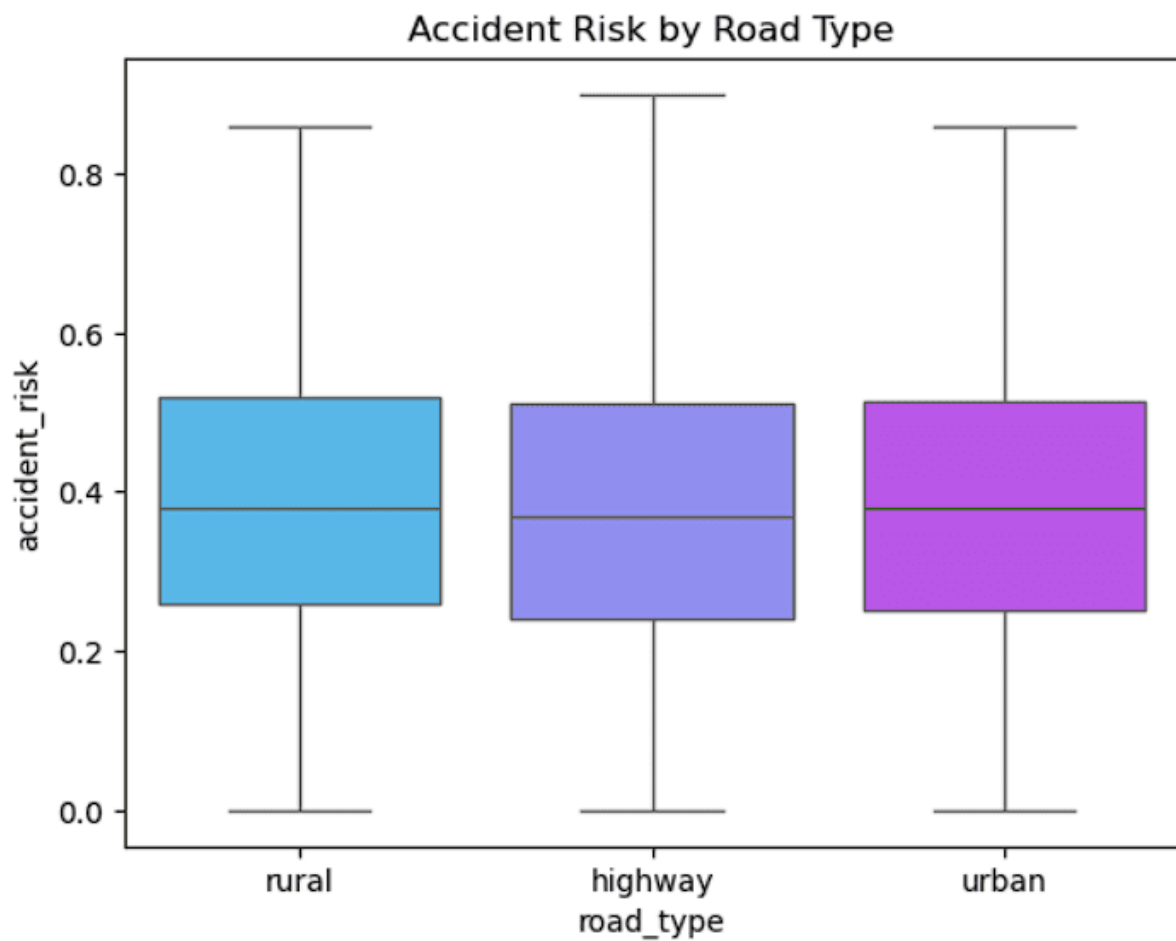
Visualization 1: Accident Risk Distribution

```
sb.histplot(df['accident_risk'], kde=True, color='crimson')  
plt.title("Distribution of Accident Risk")  
plt.show()
```



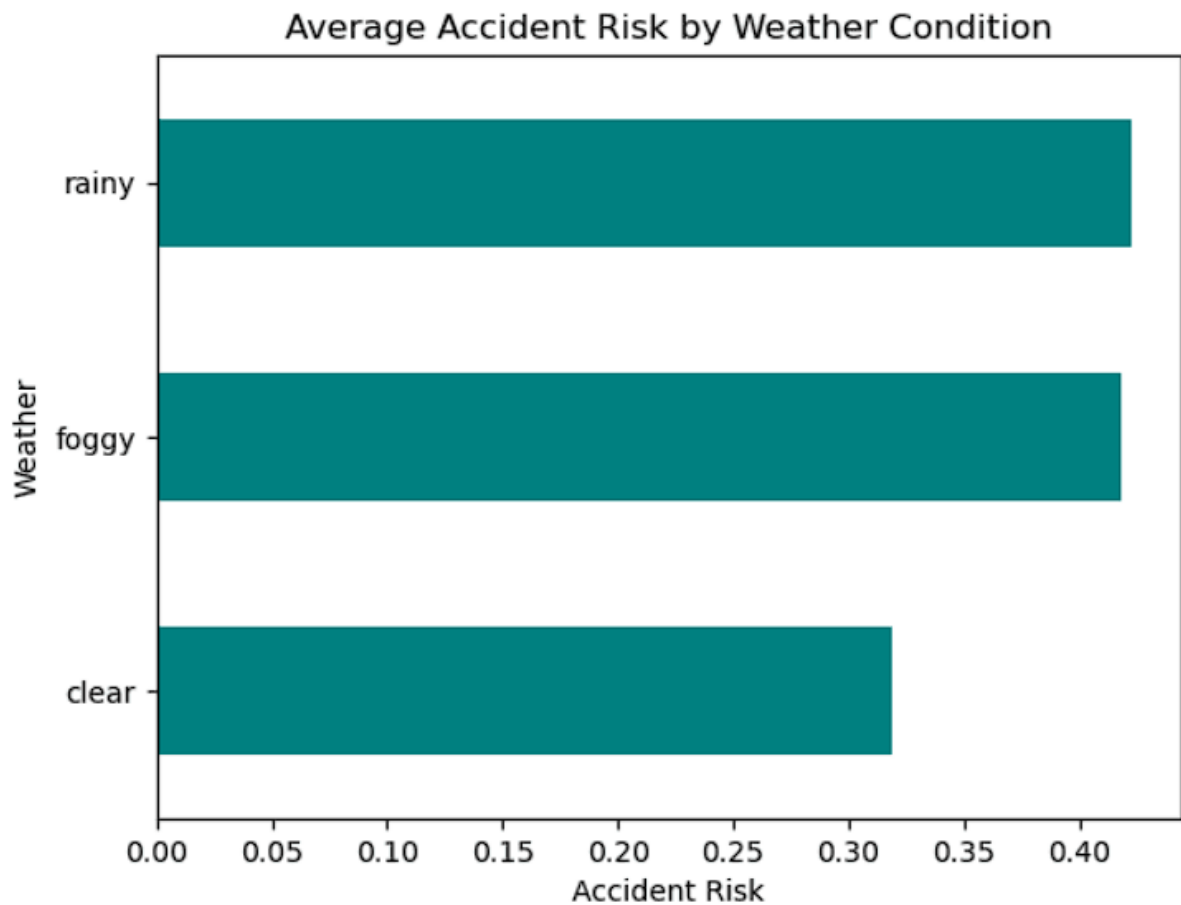
Visualization 2: Road Type vs Accident Risk

```
sb.boxplot(x='road_type', y='accident_risk', data=df, palette='cool')  
plt.title("Accident Risk by Road Type")  
plt.show()
```



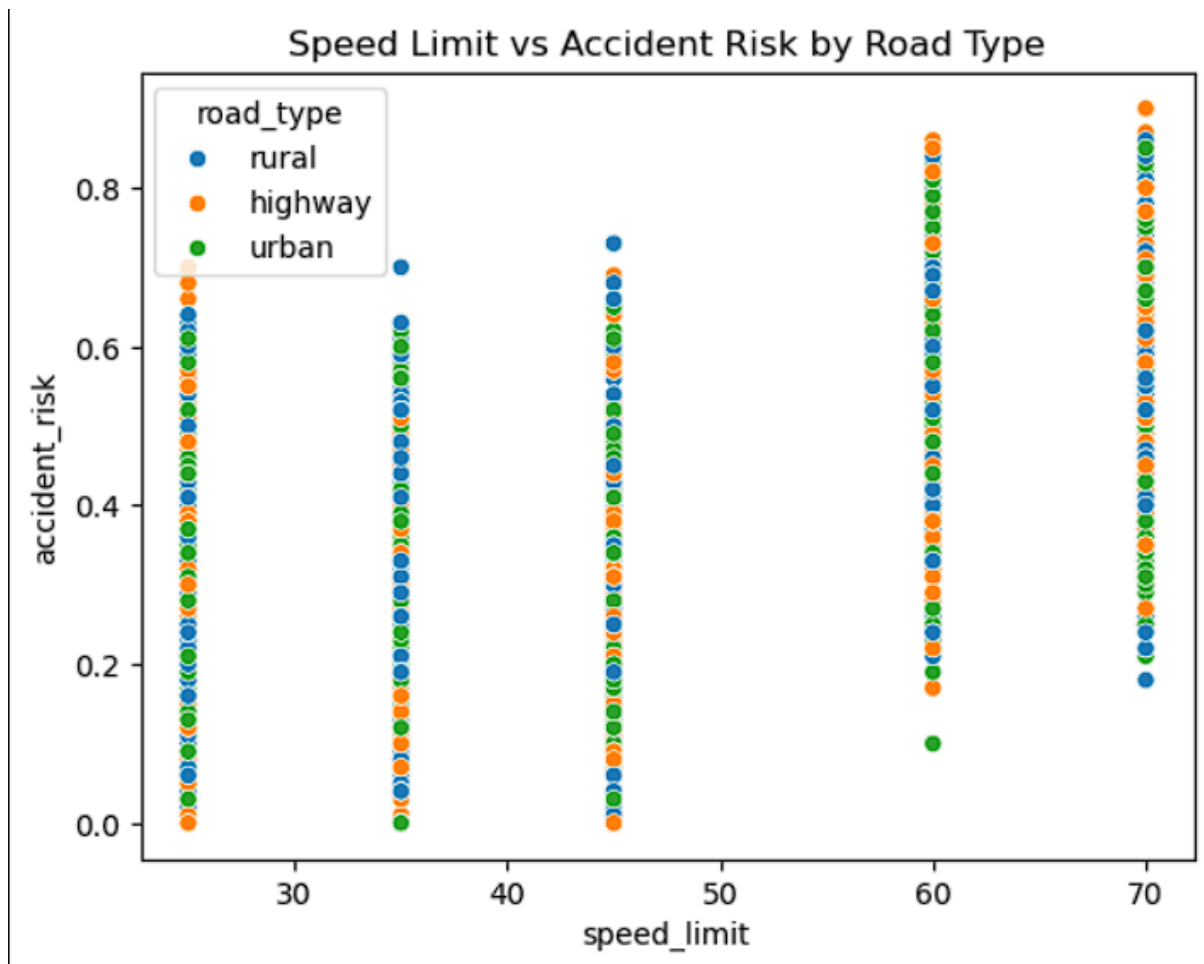
Visualization 3: Weather vs Average Risk

```
avg_risk_weather = df.groupby('weather')['accident_risk'].mean().sort_values()
avg_risk_weather.plot(kind='barh', color='teal')
plt.title("Average Accident Risk by Weather Condition")
plt.xlabel("Accident Risk")
plt.ylabel("Weather")
plt.show()
```



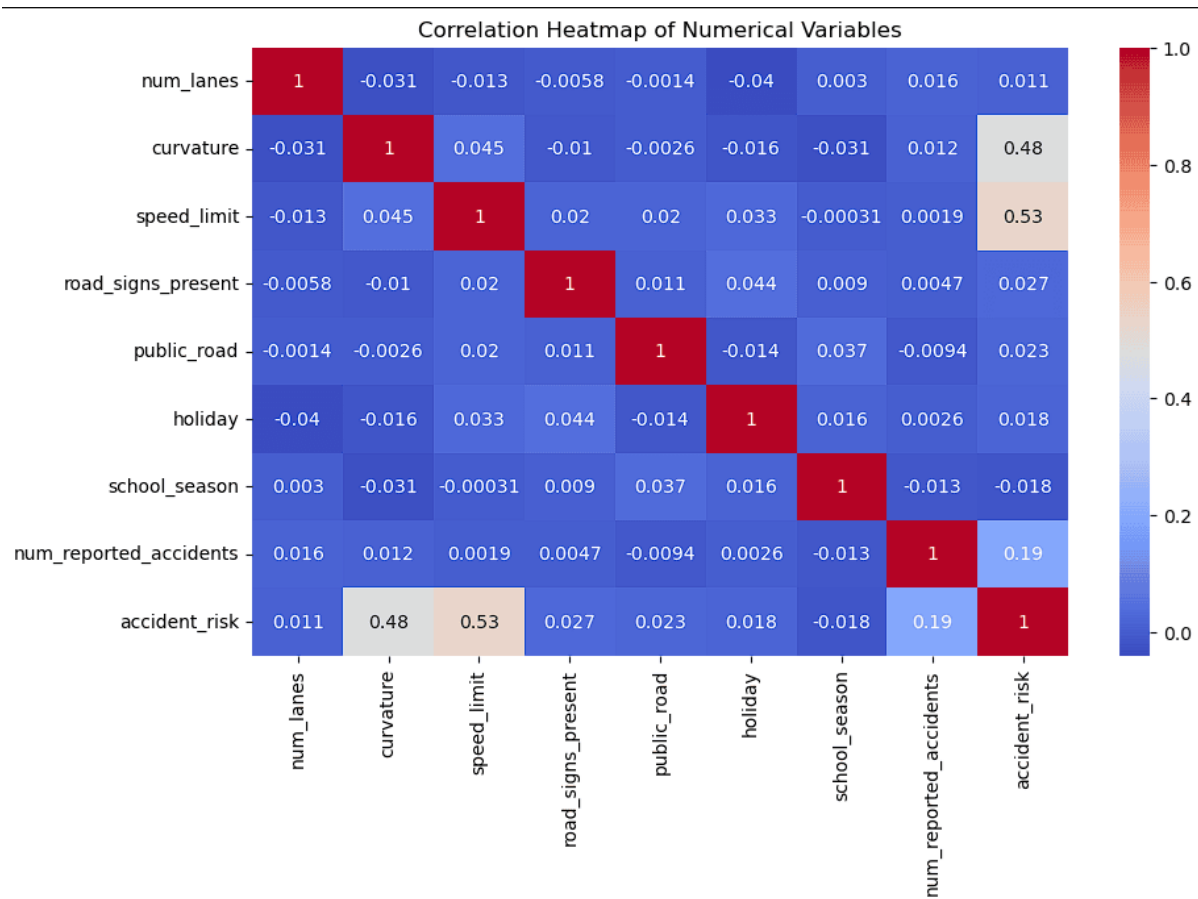
Visualization 4: Speed Limit vs Accident Risk

```
sb.scatterplot(x='speed_limit', y='accident_risk', data=df, hue='road_type')  
plt.title("Speed Limit vs Accident Risk by Road Type")  
plt.show()
```



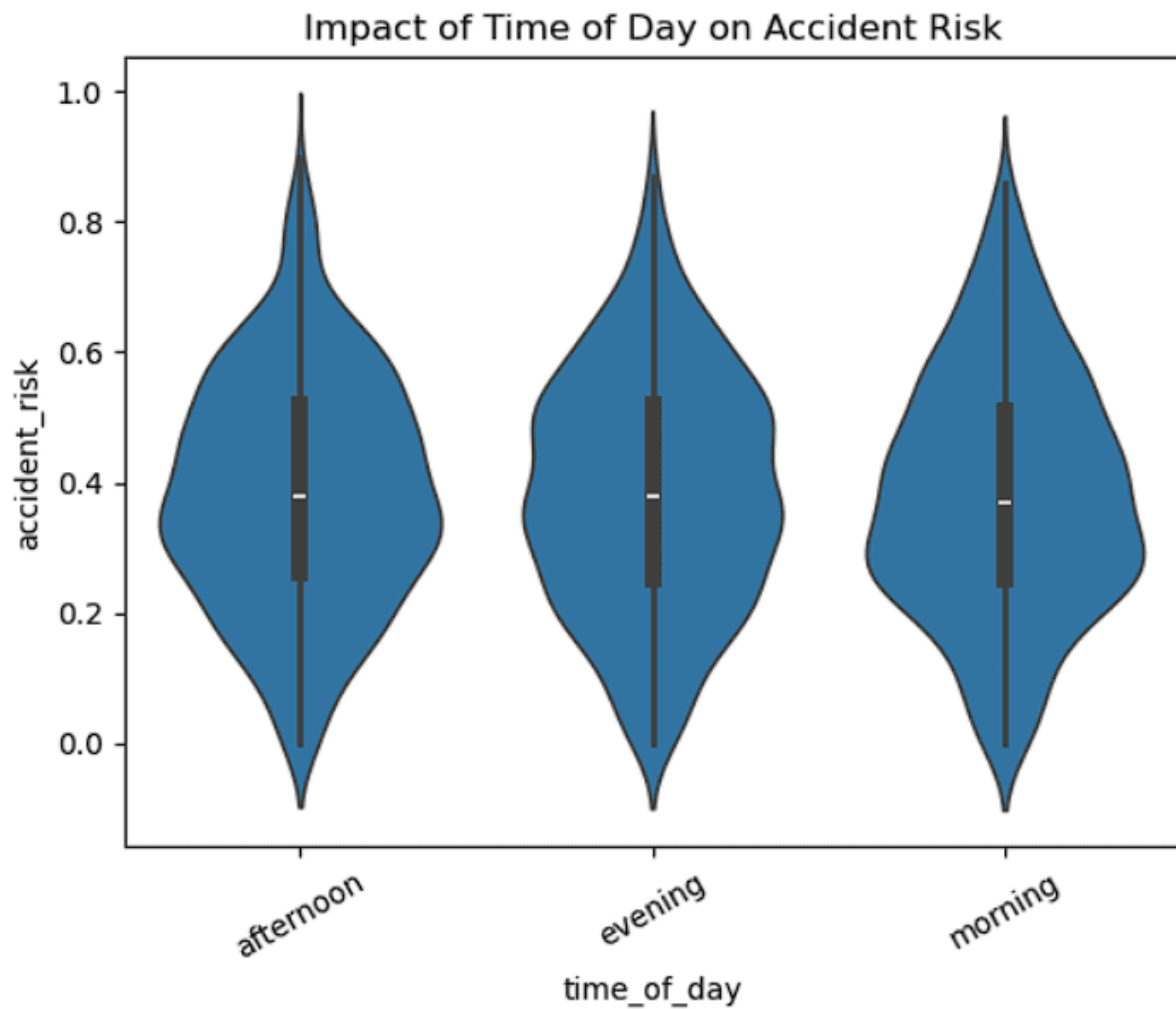
Visualization 5: Heatmap - Numeric Correlations

```
plt.figure(figsize=(10, 6))
sb.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap of Numerical Variables")
plt.show()
```



Visualization 6: Time of Day Impact

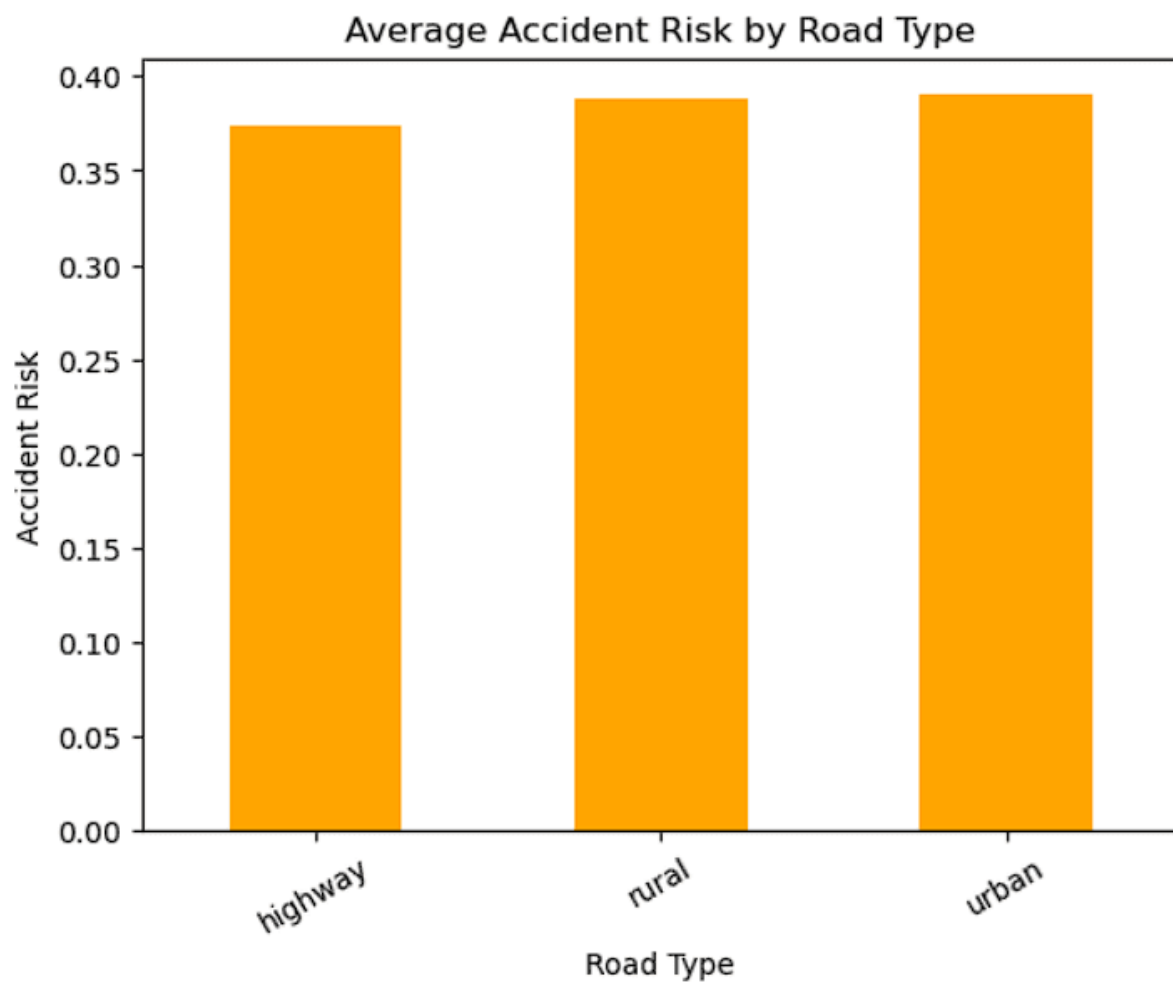
```
sb.violinplot(x='time_of_day', y='accident_risk', data=df)
plt.title("Impact of Time of Day on Accident Risk")
plt.xticks(rotation=30)
plt.show()
```



Visualization 7 : Average Risk by Road Type (Bar Chart)

```
avg_risk = df.groupby('road_type')['accident_risk'].mean().sort_values()

avg_risk.plot(kind='bar', color='orange')
plt.title("Average Accident Risk by Road Type")
plt.xlabel("Road Type")
plt.ylabel("Accident Risk")
plt.xticks(rotation=30)
plt.show()
```



KEY TAKEAWAYS:

1. Accident Risk Distribution

Most observations cluster in the **moderate range (~0.3–0.6)**, indicating that extreme low/high risks are less frequent.

2. Road Type Influence: Clear variation across road types:

- Some categories show **higher median risk and wider spread** (greater variability).
- Others are more **consistent and lower-risk**.
This suggests infrastructure and usage patterns significantly affect risk.

3. Weather Factor: Average risk differs by weather condition:

- **Adverse weather (rain, fog, etc.) shows higher average risk.**
- Clear conditions tend to have lower values.

4. Speed Limit Relationship: The scatter plot indicates a positive trend:

- Risk generally increases with speed.
- The spread widens at higher speeds, meaning **greater unpredictability**.

5. Correlations: From the heatmap:

- **Speed limit shows a positive correlation with accident risk.**
- Other numerical variables may have weaker or moderate relationships.

6. Time of Day Impact

The violin plot shows:

- **Night/dim periods have higher median risk and wider distributions.**
- Daytime conditions are relatively more stable and lower-risk.

Code Written on: VS Code

```
File Edit Selection View Go Run ... Python
Project1.py Project2.py
BCT > Project > Project1.py > ...
1 import numpy as np
2 import pandas as pd
3 import seaborn as sb
4 import matplotlib as plt
5
6 df=pd.read_csv(r"C:\Users\Debanga Mondal\Desktop\Python\BCT\Project\synthetic_road_accidents_2k.csv")
7 print(df)
8
9 print(df.head())
10
11 # Data Cleaning
12 df.dropna(inplace=True)
13
14 # Visualization 1: Accident Risk Distribution
15 sb.histplot(df['accident_risk'], kde=True, color='crimson')
16 plt.title("Distribution of Accident Risk")
17 plt.show()
18
19 # Visualization 2: Road Type vs Accident Risk
20 sb.boxplot(x='road_type', y='accident_risk', data=df, palette='cool')
21 plt.title("Accident Risk by Road Type")
22 plt.show()
23
24 # Visualization 3: Weather vs Avg Risk
25 avg_risk_weather = df.groupby('weather')['accident_risk'].mean().sort_values()
26 avg_risk_weather.plot(kind='barh', color='teal')
27 plt.title("Average Accident Risk by Weather Condition")
28 plt.xlabel("Accident Risk")
29 plt.ylabel("Weather")
30 plt.show()
31
32 # Visualization 4: Speed Limit vs Accident Risk
33 sb.scatterplot(x='speed_limit', y='accident_risk', data=df, hue='road_type')
34 plt.title("Speed Limit vs Accident Risk by Road Type")
35 plt.show()
36
37 # Visualization 5: Heatmap - Numeric Correlations
38 plt.figure(figsize=(10, 6))
39 sb.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
40 plt.title("Correlation Heatmap of Numerical Variables")
41 plt.show()
```

```
File Edit Selection View Go Run ... Python
Project1.py Project2.py
BCT > Project > Project1.py > ...
42 plt.show()
43
44 # Visualization 4: Speed Limit vs Accident Risk
45 sb.scatterplot(x='speed_limit', y='accident_risk', data=df, hue='road_type')
46 plt.title("Speed limit vs Accident Risk by Road Type")
47 plt.show()
48
49 # Visualization 5: Heatmap - Numeric Correlations
50 plt.figure(figsize=(10, 6))
51 sb.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
52 plt.title("Correlation Heatmap of Numerical Variables")
53 plt.show()
54
55 # Visualization 6: Time of Day Impact
56 sb.violinplot(x='time_of_day', y='accident_risk', data=df)
57 plt.title("Impact of Time of Day on Accident Risk")
58 plt.xticks(rotation=30)
59 plt.show()
60
61 # Visualization 7 : Average Risk by Road Type (Bar Chart)
62 avg_risk = df.groupby('road_type')['accident_risk'].mean().sort_values()
63
64 avg_risk.plot(kind='bar', color='orange')
65 plt.title("Average Accident Risk by Road Type")
66 plt.xlabel("Road Type")
67 plt.ylabel("Accident Risk")
68 plt.xticks(rotation=30)
69 plt.show()
```

Dataset:

AutoSave

<

CONCLUSION

The visual analysis of road accident data provides crucial insights into *safety-critical factors*. Using Python's data visualization tools, this study reveals:

- Highways and rural roads have greater accident risks.
- Lighting, weather, and curvature substantially contribute to higher risk scores.
- Proactive measures like improved road lighting and curvature management may effectively reduce accidents.

This project shows how Exploratory Data Analysis can enhance decision-making in transportation safety, turning raw data into meaningful visual evidence.